

CARECHAIN: A Decentralized Medical Record Exchange Using Blockchain and IPFS

MUHAMMED ASHIQUE U K

MCA S4

TCR24MCA-2042

Main Project

Guided By:

Dr. Sminesh C N

Professor & HOD

Department of Computer Applications
Government Engineering College, Thrissur, Kerala





Outline

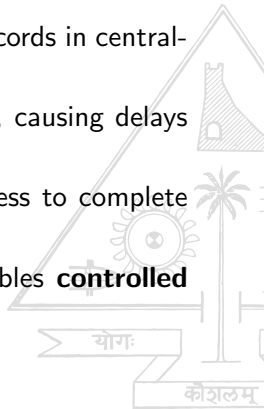
- 1 Introduction
- 2 Literature Review
- 3 Existing System
- 4 Proposed System
- 5 Requirements Specification
- 6 Input and Output
- 7 System Architecture
- 8 Data Flow Diagram
- 9 System Modules
- 10 Final Results
- 11 Performance Evaluation
- 12 Conclusion
- 13 Future Enhancements





Introduction

- Traditional healthcare systems store medical records in centralized hospital databases.
- Medical data is fragmented across institutions, causing delays and repeated diagnostics.
- Emergency care is often provided without access to complete patient history.
- There is a need for a secure system that enables **controlled and reliable medical data sharing**.





Introduction (contd.)

- **CareChain** is a blockchain-based platform for secure medical record exchange.
- Uses **blockchain technology** to ensure data integrity and transparent access logs.
- Uses **decentralized storage (IPFS)** for secure and distributed medical file storage.
- Enables consent-based sharing and emergency access across healthcare institutions.





Literature Review

Reegu et al., "Blockchain-based framework for interoperable electronic health records", 2023

- Proposes a blockchain-based framework for interoperable EHR systems.
- Highlights issues of data silos and poor portability.
- **Observation:** Validates blockchain for secure cross-institution data sharing.

Mohsan et al., "Decentralized patient-centric medical image management using blockchain and IPFS", 2022

- Presents a hybrid blockchain and IPFS-based medical data system.
- **Observation:** Supports use of IPFS for encrypted medical record storage.



Literature Review

Mandinyenya et al., “Smart contract-based consent management for secure data sharing”, 2025

- Designs a consent management model using smart contracts.
- Enables granular and purpose-based access control.
- **Observation:** Justifies smart contract-driven patient consent management.

Mishra et al., “Exploring blockchain-based patient-centric healthcare”, 2025

- Comprehensive review of patient-centric healthcare blockchain systems.
- Discusses privacy, security, and usability challenges.
- **Observation:** Reinforces need for patient-controlled healthcare platforms.



Existing System

- Healthcare records are stored in **centralized EHR systems** within individual institutions.
- Medical data is fragmented into **isolated hospital data silos**.
- Limited **patient control and consent management** over data sharing.
- Absence of transparent audit trails and standardized emergency access.





Proposed System

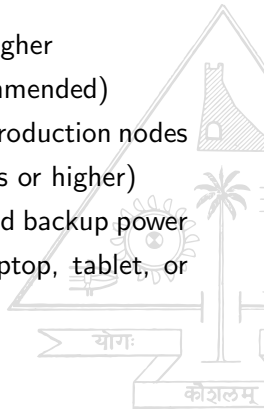
- Decentralized medical record exchange platform:
 - **Normal Access:** Consent-based data sharing across institutions.
 - **Emergency Access:** Verified break-glass access with safeguards.
- Uses **Blockchain** for immutable audit trails and access control.
- Uses **IPFS** for encrypted and distributed medical record storage.
- Smart contracts for automated consent and permission management.
- Supports interoperable healthcare data exchange standards.
- Enables secure sharing between hospitals, doctors, and insurers.





Hardware Requirements

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **Memory:** Minimum 8 GB RAM (16 GB recommended)
- **Storage:** 500 GB for development, 1 TB+ for production nodes
- **Network:** Stable internet connection (50 Mbps or higher)
- **Server Nodes:** Multi-node setup with RAID and backup power
- **Client Devices:** Any browser-enabled PC, laptop, tablet, or smartphone





Software Requirements

- **Operating System:** Windows 10/11 or Linux (Ubuntu 20.04+)
- **Blockchain Platform:** Ethereum (Ganache for development)
- **Smart Contracts:** Solidity with Truffle / Hardhat
- **Backend:** Node.js
- **Frontend:** React.js with Bootstrap
- **Storage & Tools:** IPFS, Web3.js, MetaMask





Functional Requirements

- **Patient Registration:** The system shall allow patients to register and manage digital identities and basic profile information.
- **Medical Record Management:** The system shall support secure upload of medical records with encryption and IPFS-based storage.
- **Access Control:** The system shall implement three-tier access for treatment, sharing, and emergency scenarios.
- **Consent Management:** The system shall allow patients to grant, modify, and revoke data access permissions.
- **Audit and Emergency Access:** The system shall maintain access logs and provide verified emergency break-glass access.



Non-Functional Requirements

- **Security:** The system shall ensure secure data handling using encryption, role-based access control, and immutable audit logs.
- **Performance:** The system shall support fast medical record upload and retrieval with acceptable response times under normal load.
- **Reliability:** The system shall provide high availability and data durability using distributed storage and blockchain mechanisms.
- **Usability:** The system shall offer intuitive interfaces accessible to patients and healthcare providers with minimal training.
- **Scalability & Compliance:** The architecture shall scale across institutions while supporting healthcare standards and regulatory requirements.



Input and Output

Input

- **Medical Records:** Uploaded reports, images, and prescriptions.
- **Patient Data:** Registration details and consent preferences.
- **Access Requests:** Requests from doctors or hospitals to view records.

Output

- **Secure Record Access:** Authorized viewing of decrypted medical data.
- **Consent & Audit Logs:** Patient-visible access history and permissions.
- **Notifications:** Alerts for emergency access and system activities.



System Architecture

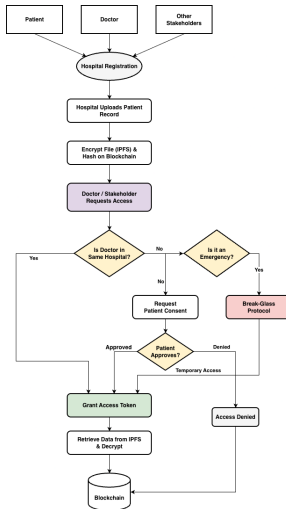


Figure: Architecture Diagram



Data Flow Diagrams

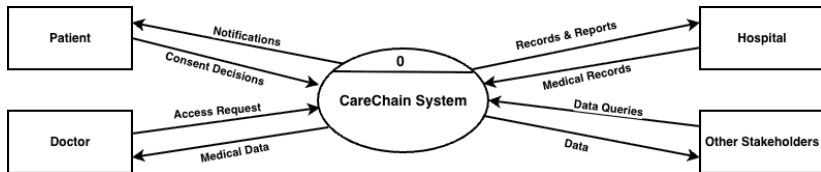


Figure: Level 0 DFD



Data Flow Diagrams

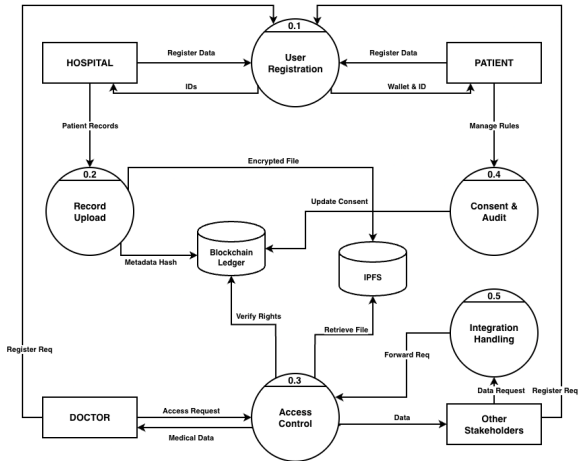


Figure: Level 1 DFD



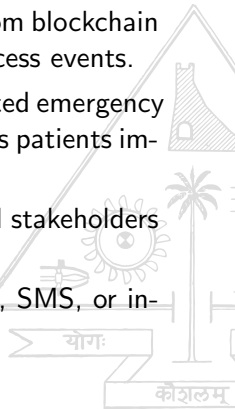
System Modules

- **User Registration Manager:** Handles user onboarding, wallet creation, and role-based registration for patients, doctors, hospitals, and stakeholders.
- **Record Upload & Encryption Handler:** Encrypts medical files using AES-256, stores them on IPFS, and records metadata on the blockchain.
- **Access Control Engine:** Implements three-tier access (same-hospital, consent-based, emergency) by verifying permissions on blockchain and retrieving files from IPFS.
- **Consent Management System:** Allows patients to define granular consent rules, which are stored immutably on the blockchain.



System Modules

- **Audit Trail Manager:** Generates audit reports from blockchain logs and sends real-time notifications for data access events.
- **Emergency Protocol Handler:** Grants time-limited emergency access with multi-factor authentication and notifies patients immediately.
- **Integration Gateway:** Provides APIs for external stakeholders with anonymization and compliance enforcement.
- **Notification Dispatcher:** Sends alerts via email, SMS, or in-app for consent changes and access events.





Implementation

- Successfully implemented all core modules including **Smart Contract Development, Backend API, Frontend Dashboards,** and **IPFS Integration**.
- Smart contracts written in **Solidity** using **Hardhat** framework, deployed on **Ethereum private network (Ganache)** with **three-tier access logic** and **immutable audit logs**.
- Backend built with **Node.js**, handling **AES-256 encryption** of medical files, **IPFS uploads, temporary key management,** and **event polling** for access grants.
- Frontend developed in **React** with separate dashboards for Patients, Doctors, and Hospital Admins, integrated with **Web3.js** and **MetaMask** for wallet-based authentication.



Implementation

- **Three-tier access control: Tier 1** (same-hospital auto-access), **Tier 2** (patient consent via smart contract), and **Tier 3** (emergency break-glass with **24-hour time limit**).
- **Emergency access protocol** triggers automatic **patient notification** via email/in-app alerts and logs all attempts **immutably on-chain**.
- Medical records encrypted with **AES-256** before **IPFS storage**; decryption keys stored temporarily in backend and released only to **authorized doctors**.
- **Audit trail** feature allows patients to view complete access history including **doctor address**, **timestamp**, and **access tier**.



Final Results

Record uploaded successfully. Record ID: 3

Hospital: H1 (ID: 1)

Patients registered with this hospital:

Patient Address	Action
0x87740C827a5D8BedB362340f424a9c1623248F21	<button>Selected</button>

Upload Medical Record

Selected Patient

0x87740C827a5D8BedB362340f424a9c1623248F21

Medical Record File

Choose File Screenshot 2026-02-19 at 11:29:49 AM

UPLOAD RECORD

Existing Records for 0x87740C827a5D8BedB362340f424a9c1623248F21

ID	Hospital ID	Timestamp	IPFS Hash	Action
1	1	2/26/2026, 3:47:51 PM	QmIWQ44t6D...	<button>DOWNLOAD</button>

Figure: Hospital Dashboard



Final Results

Patient Dashboard

Connected as: 0x01985a3d72febe3593f42043fec74c3e02b0cbbe

My Records

Manage Access

LOAD MY RECORDS

ID	Hospital	Timestamp	IPFS Hash	Logs	Download
4	2	2/26/2026, 8:03:34 PM	QmU8Hw8sZG...	VIEW LOGS	DOWNLOAD

Access Logs

Doctor Address	Doctor Name	Hospital	Tier	Time
0xE7b9858f4FD27F374ed31dB98f9Ad6414D5bD73	Jhon	H1	3	2/26/2026, 8:06:54 PM

Figure: Patient Dashboard



Final Results

Doctor Dashboard

Connected as: 0xe7b9858f4ffd271374ed31db98f9ad6414d5bd73

Hospital ID: 1

My Hospital Patients

Other Hospitals

Patients at your hospital

Patient Address	Action
0x87740C827a5D8BedB362340f424a9c1623248F21	<button>Selected</button>

Patient Details

Name: Ashok

Age: 44

Gender: Male

Records for 0x87740C827a5D8BedB362340f424a9c1623248F21

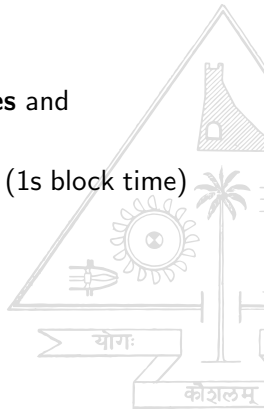
ID	Hospital	Timestamp	IPFS Hash	Action
1	1	2/26/2026, 3:47:51 PM	QmWQ144t6D...	<button>DOWNLOAD</button>
2	1	2/26/2026, 4:41:38 PM	QmREVUKTsf...	<button>DOWNLOAD</button>
3	1	2/26/2026, 7:56:28 PM	QmNuYPux9C...	<button>DOWNLOAD</button>

Figure: Doctor Dashboard



Performance Evaluation

- Evaluated on: **transaction confirmation times** and **encryption/decryption speed**.
- Tests performed on local **Ganache blockchain** (1s block time) and **Node.js backend**.

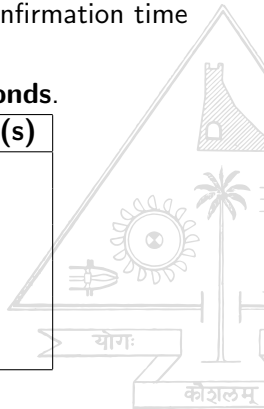




Transaction Processing Times

- Each operation executed **10 times**; average confirmation time recorded.
- All transactions confirmed within **1.0–1.3 seconds**.

Operation	Avg Time (s)
User Registration	1.2
Hospital Registration	1.3
Record Upload (meta)	1.1
Consent Granting	1.0
Access Request (T1/T2)	1.1
Emergency Access	1.2



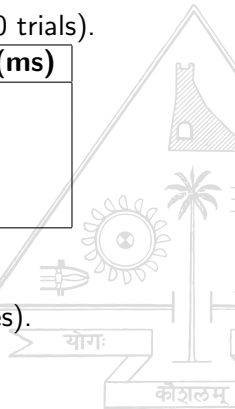


Encryption / Decryption Performance

AES-256-CBC encryption on files of varying sizes (10 trials).

File Size	Encryption (ms)	Decryption (ms)
100 KB	45	38
1 MB	210	195
5 MB	890	820
10 MB	1720	1650

- Even **10 MB files process in under 2 seconds.**
- Suitable for typical medical reports (PDFs, images).

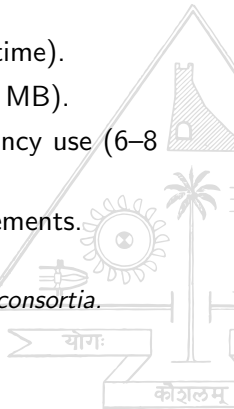




Summary of Performance

- **Blockchain latency** – negligible (≈ 1 sec block time).
- **Encryption overhead** – minimal (< 2 sec for 10 MB).
- **End-to-end access** – acceptable for non-emergency use (6–8 sec).
- System meets **real-world responsiveness** requirements.

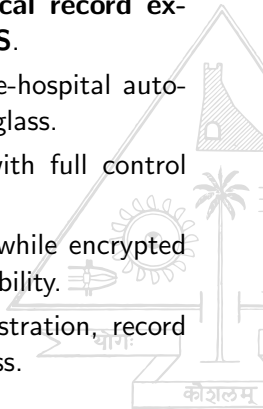
CareChain is ready for pilot deployment in healthcare consortia.





Conclusion

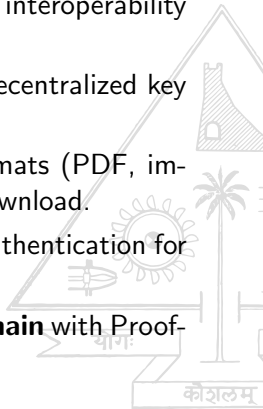
- Successfully developed a **decentralized medical record exchange** using **Ethereum blockchain** and **IPFS**.
- Implemented **three-tier access control**: same-hospital auto-access, patient consent, and emergency break-glass.
- **Patient-centric design** empowers patients with full control over consent and transparent audit logs.
- Hybrid architecture stores metadata on-chain while encrypted files reside on IPFS, ensuring privacy and scalability.
- System validated for core functionalities: registration, record upload, consent management, and secure access.





Future Enhancements

- Integrate **HL7 FHIR standards** for seamless interoperability with existing hospital EHR systems.
- Implement **attribute-based encryption** and decentralized key management for enhanced end-to-end privacy.
- Add **in-browser file preview** for common formats (PDF, images) to improve usability without requiring download.
- Develop **mobile applications** with biometric authentication for patients and doctors.
- Deploy on a **permissioned consortium blockchain** with Proof-of-Authority for higher throughput.





References

- F. A. Reegu, H. Abas, Y. Gulzar, Q. Xin, A. A. Alwan, A. Jabbari, R. G. Sonkamble, and R. A. Dziyauddin, “Blockchain-based framework for interoperable electronic health records for an improved healthcare system,” *Sustainability*, vol. 15, no. 8, p. 6337, 2023.
- S. A. H. Mohsan, A. Razzaq, S. A. K. Ghayyur, H. K. Al-kahtani, N. Al-Kahtani, and S. M. Mostafa, “Decentralized patient-centric report and medical image management system based on blockchain technology and the IPFS,” *International Journal of Environmental Research and Public Health*, vol. 19, no. 22, p. 14641, 2022.
- M. Rasel, R. Bommur, R. B. Shovon, and M. A. Islam, “Blockchain-enabled secure interoperability: advancing electronic health records data exchange,” *International Journal of Advanced Engineering Technologies and Innovations*, vol. 1, no. 2, pp. 193–211, 2022.



References

- A. Shahzad, W. Chen, Y. Zhang, and R. Kumar, “Zero-trust medical image sharing: a secure and decentralized approach using blockchain and the IPFS,” *Symmetry*, vol. 17, no. 4, 2025.
- G. Mandinyenya and V. Malele, “Design and implementation of a smart contract-based consent management model for secure personal data sharing,” *Jurnal Ilmiah Computer Science*, vol. 4, no. 1, pp. 23–38, 2025.
- D. K. Mishra and P. S. Mehra, “Exploring blockchain-based patient-centric healthcare: a comprehensive review,” *Concurrency and Computation: Practice and Experience*, vol. 37, no. 12–14, p. e70124, 2025.



Thank You

